

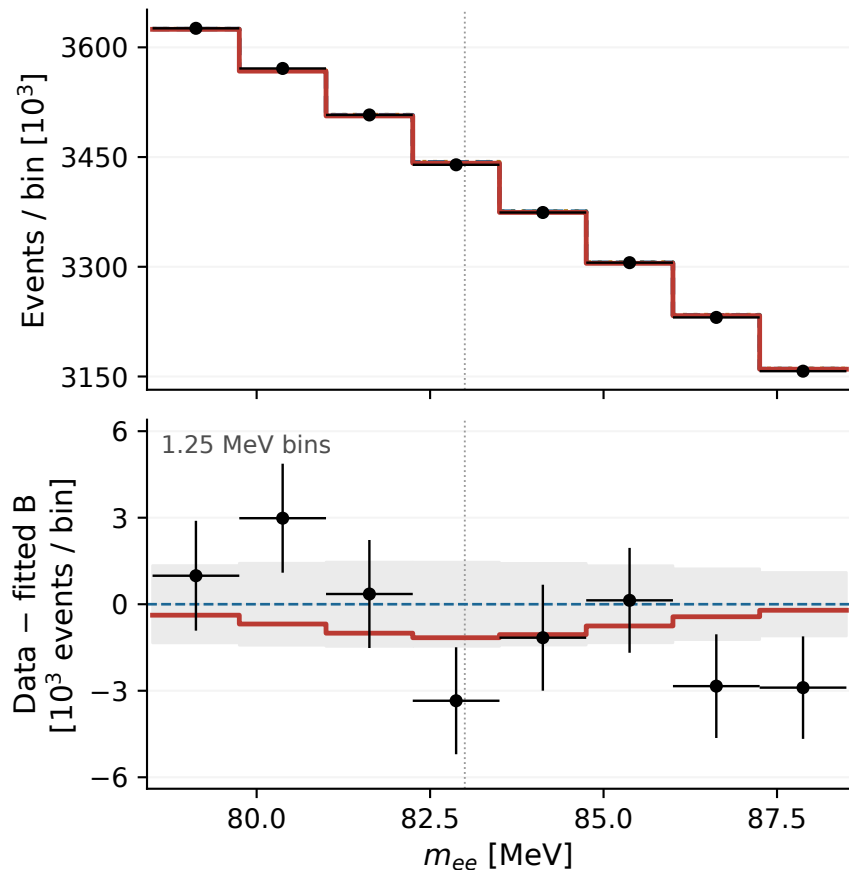
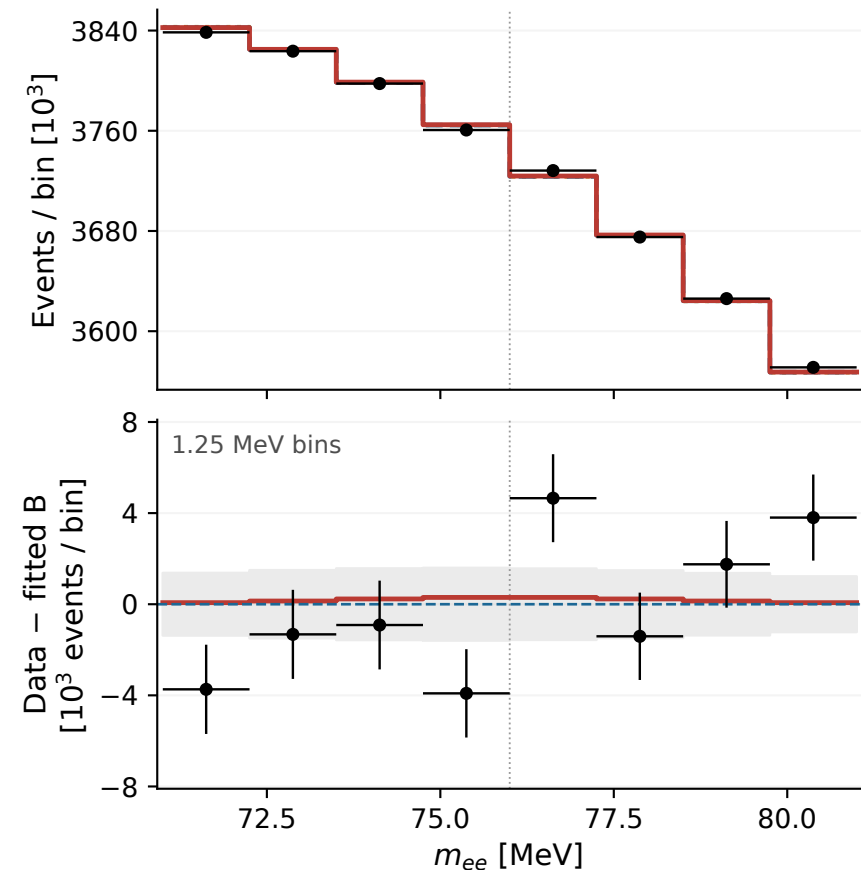
Extreme stress-centered tails can have small fitted signals

76 MeV: stress offset $a = -8.70$ | 83 MeV: stress offset $a = +7.71$

- Observed data
- Background fitted with signal
- GP constraint SD (width guide)
- ⋯ GP mean before profiling
- S+B (top); fitted S (bottom)

76 MeV: observed $r = +0.17$

83 MeV: observed $r = -0.68$



Centering changes the probability assigned to a fit. It does not turn the small red template into a larger fitted signal.

Bars: counting error only. Gray shading: zero-centered GP constraint width; not fitted-background error or total residual uncertainty.